⇒ Markov Chain Monte Carlo

⇒ Goal : Draw : $\theta_1, \theta_2, \cdots \theta_n \sim f$

⇒ theory of Markov Chains

$\theta_1 \quad \theta_2 \quad \theta_3 \quad \cdots$

$\{\theta_i\}$ is a random process

⇒ Definition:

$\{\theta_i \mid i = 1, 2, \cdots\}$ is called a Markov chain if

$$P(\theta_i \mid \theta_1, \theta_2, \cdots \theta_{i-1})$$

$$= P(\theta_i \mid \theta_{i-1})$$

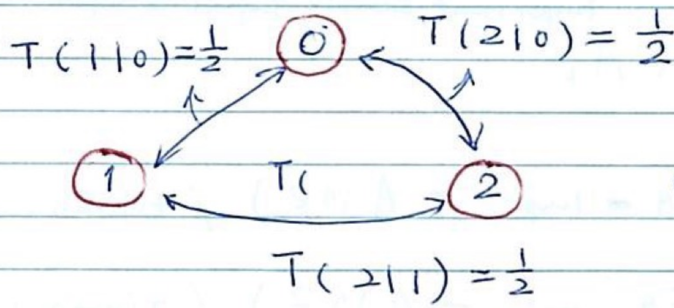
We denote $P(\theta_i \mid \theta_{i-1})$ by $T(\theta_i \mid \theta_{i-1})$ and call it transition distribution

$$\theta_0 \xrightarrow{T(\theta_1 \mid \theta_0)} \theta_1 \xrightarrow{T(\theta_2 \mid \theta_1)} \theta_2$$

Example: state space $= \{0, 1, 2, \}$

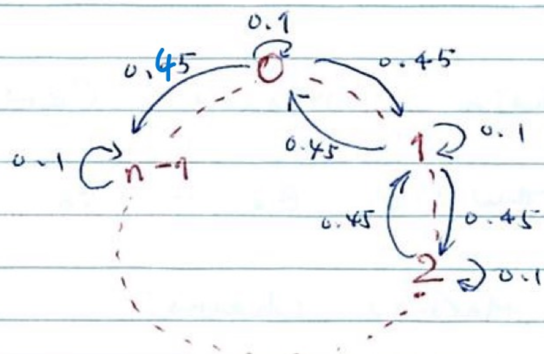
$T \longrightarrow \pi$

$T \longleftarrow \pi$



$0 \rightarrow 1 \rightarrow 2$

example:



$\Rightarrow$ a function that simulation a

Markov chain

⇒ Convergence of Markov chain

under some regularity conditions.

for $\forall\ \theta_0$, we have

$$\lim_{n\to\infty} P(\theta_n = \theta \mid \theta_0 = \theta_0) = \pi(\theta)$$

The $\pi(\theta)$ is called the equilibrium Distribution

of the Markov chain.

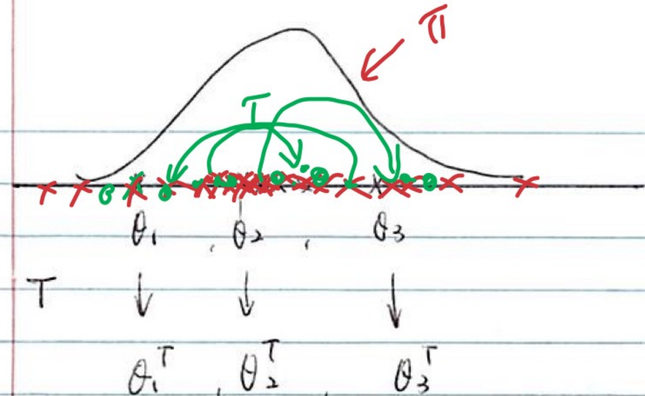
⇒ Invariance condition:

$\pi(\theta)$ is a distribution; $T(\cdot,\cdot)$ is a

Markov chain transition. We say

"$T(\cdot,\cdot)$ leaves $\pi(\theta)$ invariant" if

$$\int \pi(\theta)\, T(\theta, \theta^T)\, d\theta = \pi(\theta^T)$$



If $\theta \sim \pi$, $\theta^T \sim T(\theta, \cdot)$, then $\theta^T \sim \pi$

Theorem:

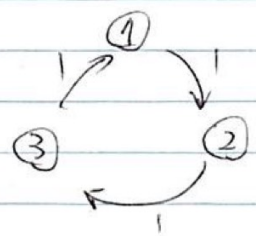
If a marko chain has an esaibirium distribution π, then $\int \pi(\theta)\, T(\theta, \theta^T)\, d\theta = \pi(\theta^T)$

Markov-Chain Monte Carlo

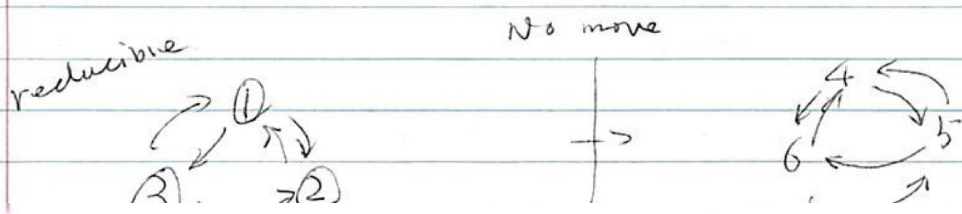
Given a target distribution π, we design a markov chain $T(\cdot,\cdot)$ such that

1) Invariance condition

2) $T(\cdot,\cdot)$ is aperiodic



3) $T(\cdot,\cdot)$ is irreducible



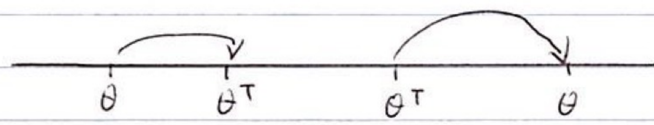
then, we can simulate T long enough, after we discard the initial value, we can view the remaining steps have distribution π.

$\Rightarrow$ Detailed balance condition

reversibility cond.

$$\pi(\theta)\, T(\theta, \theta^T) = \pi(\theta^T)\, T(\theta^T, \theta) \quad \text{for all } \theta,$$

θ^T, also called "reverisible"



NOT: $T(\theta, \theta^T) = T(\theta^T, \theta)$

$\Rightarrow$ Theorem :

Detailed balance Condition $\Rightarrow$ invariance conditio

proof: $\int \pi(\theta)\, T(\theta, \theta^T)\, d\theta \overset{?}{=} \pi(\theta^T)$

$= \pi(\theta^T) \int T(\theta^T, \theta)\, d\theta$ $\{$ where, $\int T(\theta^T, \theta)\, d\theta = 1$ $\}$

$= \pi(\theta^T)$

⇒ Gibbs sampler:

We want to draw $(\theta_1, \theta_2) \sim \pi(\theta_1, \theta_2)$

Gibbs sampler:

starting from $(\theta_1^{(0)}, \theta_2^{(0)})$

$\pi(\theta_1 \mid \underline{\theta^{(0)}})$ ✓

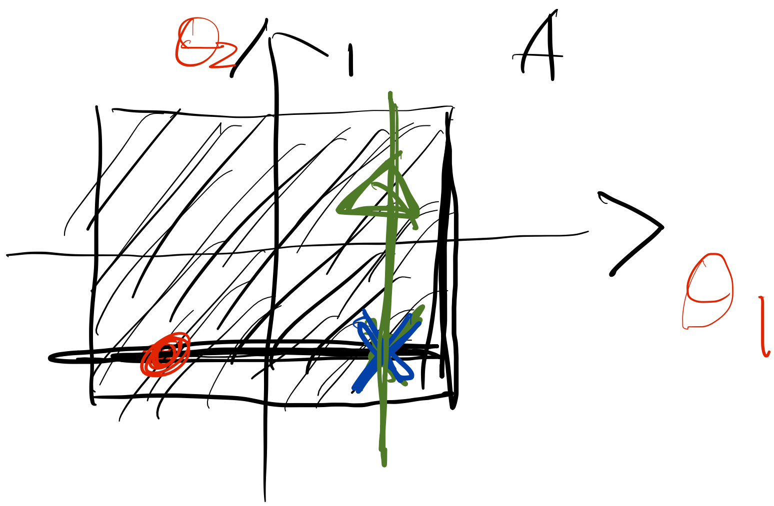
Repeat n times

1) Draw $\theta_1^{(t)} \sim \pi(\theta_1 \mid \theta_2^{(t-1)})$
2) Draw $\theta_2^{(t)} \sim \pi(\theta_2 \mid \theta_1^{(t-1)})$

A toy example:

$$(\theta_1, \theta_2) \sim unif(A)$$

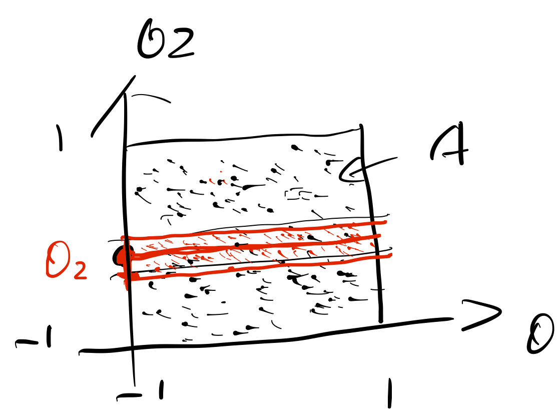
$$f(\theta_1, \theta_2) = I((\theta_1, \theta_2) \in A)$$



$$f(\theta_2) = I(-1 < \theta_2 < 1)$$

$$f(\theta_1 \mid \theta_2) = \frac{f(\theta_1, \theta_2)}{\int f(\theta_1, \theta_2) d\theta_1}$$

$= 1, \quad$ for $-1 < \theta_1 < 1$



⇒ Justification of Gibbs sampling

$(\theta_1, \theta_2) \sim \pi(\theta_1, \theta_2)$

Repeat n times

1) $\theta_1^{(t)} \sim \pi(\theta_1 \mid \theta_2^{(t-1)})$

2) $\theta_2^{(t)} \sim \pi(\theta_2 \mid \theta_1^{(t-1)})$

$T\big((\theta_1, \theta_2) \to (\theta_1^T, \theta_2^T)\big)$

$= I(\theta_1 = \theta_1^T)\,\pi(\theta_2^T \mid \theta_1)$

The left hand of D.B.E $\left\{\begin{array}{l}\text{Distribution}\\ \text{Balance}\\ \text{equation}\end{array}\right.$

$= \pi(\theta_1, \theta_2)\,\pi(\theta_2^T \mid \theta_2)\,I(\theta_1 = \theta_1^T)$

$= \pi(\theta_1)\,\pi(\theta_2 \mid \theta_1)\,\pi(\theta_2^T \mid \theta_1)\,I(\theta_1 = \theta_1^T)$

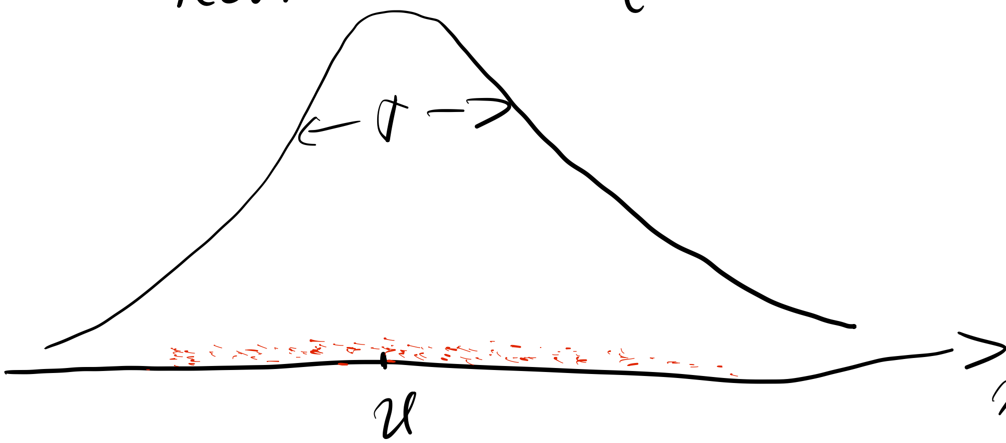
The Right hand side of D.B.E

$= \pi(\theta_1^T, \theta_2^T)\,\pi(\theta_2 \mid \theta_1^T)\,I(\theta_1 = \theta_1^T)$ → Fixed

$= \pi(\theta_1^T)\,\pi(\theta_2^T \mid \theta_1^T)\,\pi(\theta_2 \mid \theta_1^T)$

We can see that L.H.S = R.H.S

Example: Sampling from the posterior of normal sample



$$X_1, X_2, \cdots, X_n \mid (\mu, \sigma^2) \overset{iid}{\sim} N(\mu, \sigma^2)$$

$$\mu \sim N(\mu_0, \sigma_0^2)$$

$$\sigma^2 \sim IG \text{ (inverse Garmma) } \left(\frac{\alpha}{2}, \frac{\alpha\omega}{2}\right)$$

(we let $a_0 = \alpha/2$, $\lambda_0 = \alpha\omega/2$)

$$\left[\frac{1}{\sigma^2} \sim G \left(a_0 = \alpha/2, \lambda_0 = \alpha\omega/2\right) \right]$$

$$P(\sigma^2) \propto \frac{1}{(\sigma^2)^{a_0 - 1}} \cdot e^{-\lambda_0/\sigma^2} \cdot \left[\frac{1}{(\sigma^2)^2}\right]$$

$$= \frac{1}{(\sigma^2)^{a_0+1}} \cdot e^{-\lambda_0/\sigma^2}$$

where, $\begin{cases} a_0 \to \text{sape parameters} \\ \lambda_0 \to \text{scale parameters} \end{cases}$

$$P(\mu) \propto e^{-(\mu - \mu_0)^2 / 2\sigma_0^2}$$

$$IG: \quad f(x) \propto \frac{1}{x^{\alpha+1}} e^{-\frac{\lambda}{x}}$$

$\Rightarrow$ Likelihood :

$$L(\mu, \sigma^2 ; \underset{\sim}{x}) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{-(x_i-\mu)^2/2\sigma^2}$$

$$\Rightarrow L(\mu, \sigma^2 ; \underset{\sim}{x}) \propto (\sigma^2)^{-\frac{n}{2}} e^{-\sum_{i=1}^{n} \left(\frac{(x_i-\mu)^2}{2\sigma^2} \right)}$$

$$\Rightarrow P(\mu, \sigma^2 \mid \underset{\sim}{x}) \propto P(\mu) \cdot P(\sigma^2) \, L(\mu, \sigma^2 ; \bar{x})$$

$$\propto e^{-\frac{(\mu-\mu_0)^2}{2\sigma_0^2}} \cdot (\sigma^2)^{-(a_0+1)} e^{-\lambda_0/\sigma^2}$$

$$\times (\sigma^2)^{-\frac{n}{2}} e^{-\sum_{i=1}^{n} (x_i-\mu)^2/2\sigma^2}$$

Derivation of Conditional Distributions for Gibbs Sampling

To apply Gibbs samplings, we need to find the conditional distrib. of θ and σ^2

those are: $P(\mu \mid \sigma^2, \underset{\sim}{x})$ and

$$P(\sigma^2 \mid \mu, \underset{\sim}{x})$$

* { Note that $\boxed{f(x, y) = f(x \mid y) \cdot f_y(y)}$

$$\Rightarrow f(x \mid y) = f(x, y) \Big/ \int f(x, y)\, dx$$ }

$$\sum_{i=1}^{n}(\mu-x_i)^2=\sum_{i=1}^{n}(\mu-\bar{x}+\bar{x}-x_i)^2=\sum_{i=1}^{n}(\mu-\bar{x})^2+\sum_{i=1}^{n}(\bar{x}-x_i)$$

$$P(\mu \mid \sigma^2, \underset{\sim}{x}) \propto e^{-(\mu-\mu_0)^2/2\sigma_0^2} \cdot e^{-\sum_{i=1}^{n}(x_i-\mu)^2/2\sigma^2}$$

$$\propto e^{-\frac{\mu^2-2\mu\mu_0}{2\sigma_0^2}} \cdot e^{-\frac{n\mu^2-2n\mu\cdot\bar{x}}{2\sigma^2}}$$

$$= e^{-\frac{1}{2}\left(\left(\frac{1}{\sigma_0^2}+\frac{n}{\sigma^2}\right)\cdot\mu^2-2\left(\frac{\mu_0}{\sigma_0^2}+n\cdot\bar{x}/\sigma^2\right)\mu\right)}$$

(where, $\bar{x} = \sum x_i / n$)

$$\because f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$= \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x^2}{\sigma^2}-2\cdot\frac{\mu}{\sigma^2}\cdot x+\frac{\mu^2}{\sigma^2}\right)}$$

Note that: $(\mu \mid \sigma^2, \underset{\sim}{x})$ is Normal distribution

And $\frac{1}{\sigma_0^2}+\frac{n}{\sigma^2}=\frac{1}{\sigma^2_{\mu\mid\sigma^2,\underset{\sim}{x}}}$ and

$$\frac{\mu_{\mu\mid\sigma^2,\underset{\sim}{x}}}{\sigma^2_{\mu\mid\sigma^2,\underset{\sim}{x}}} = \frac{\mu_0}{\sigma_0^2}+\frac{n\bar{x}}{\sigma^2} \quad \left(\approx \frac{\sigma^2}{n}\right)$$

variable

$$E(\mu_{\mu\mid\sigma^2,\underset{\sim}{x}}) = \frac{\frac{\mu_0}{\sigma_0^2}+\frac{n\bar{x}}{\sigma^2}}{\frac{1}{\sigma_0^2}+\frac{n}{\sigma^2}} \quad (n \to \infty) \quad (\approx \bar{x})$$

n/σ^2 is called weight for data

$$Var(\mu \mid \sigma^2, x) = \sigma^2_{\mu\mid\sigma^2,\underset{\sim}{x}} = \left(\frac{1}{\sigma_0^2}+\frac{n}{\sigma^2}\right)^{-1} \quad \left(\approx \frac{\sigma^2}{n}\right)$$

$$P(\sigma^2 \mid \mu, \underset{\sim}{x}) \propto (\sigma^2)^{-(\frac{n}{2}+a_0+1)} \cdot e^{-\frac{\overset{\text{prior}}{\lambda_0} + \overset{L}{\frac{\sum (x_i-\mu)^2}{2}}}{\sigma^2}}$$

$$\sigma^2 \mid \mu, \underset{\sim}{x} \sim IG\left(\frac{n}{2}+a_0, \frac{\sum (x_i-\mu)^2}{2} + \lambda_0\right)$$

$$= IG\left(\frac{n+\alpha}{2}, \frac{\alpha w + \sum (x_i-\mu)^2}{2}\right)$$

$$a_0 = \frac{\alpha}{2}, \quad \lambda_0 = \frac{\alpha w}{2}$$

Gibbs sampling for latent variable models

"About Assignment 2"

Example: $(\underset{\sim}{X} = (X_1, \cdots X_n) \mid \mu, \sigma^2) \sim N(\mu, \sigma^2)$

$$y_i = \text{floor}(x_i)$$

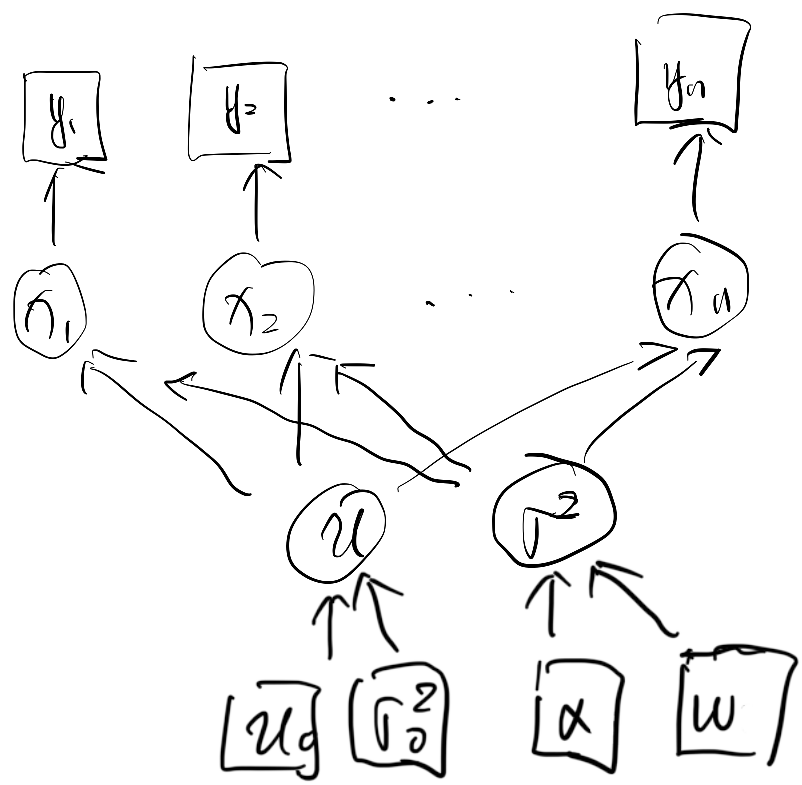
$$\mu \sim N(\mu_0, \sigma_0^2)$$

$$\sigma^2 \sim IG\left(\frac{\alpha}{2}, \alpha w/_2\right)$$

$P(y, x, \mu, \sigma^2) \quad (i = 1, 2, \cdots n)$

$= \pi(\mu) \cdot \pi(\sigma^2) \prod_{i=1}^{n} P(x_i \mid \mu, \sigma^2)$

$\prod_{i=1}^{n} \{ I(y_i = \text{floor}(x_i)) \}$ → given



observed.

← No parameters

latent

Gibbs sampling :

Repeat :

1) $x_i^{(t)} \sim x_i \mid \mu^{(t-1)}, (\sigma^2)^{(t-1)}, y_i$

2) $\mu^{(t)} \sim \mu \mid x_i^{(t)}, (\sigma^2)^{(t-1)}, y_i$

3) $(\sigma^2)^{(t)} \sim \sigma^2 \mid x_i^{(t)}, \mu^{(t)}, y_i$

To find 1)

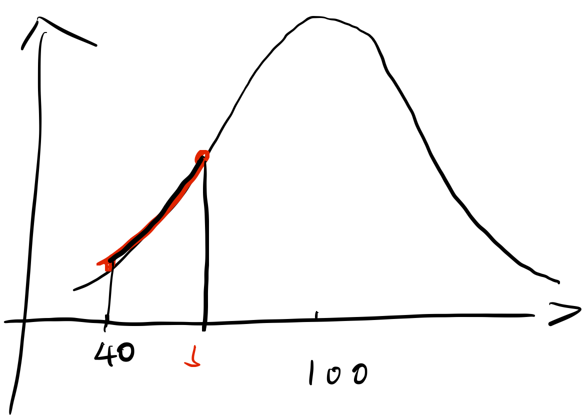
$$f(x_i \mid \mu, \sigma^2, y_i)$$

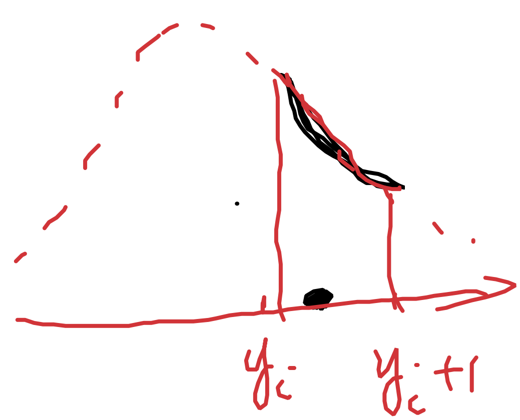
$$= \frac{p(x_i \mid \mu, \sigma^2) \cdot I(y_i = (\text{floor}(x_i)))}{\int p(x_i \mid \mu, \sigma^2) I(y_i = \text{floor}(x_i))\, dx_i}$$

$$= \frac{p(x_i \mid \mu, \sigma^2)}{\int_{y_i}^{y_i+1} p(x_i \mid \mu, \sigma^2)\, dx_i}$$

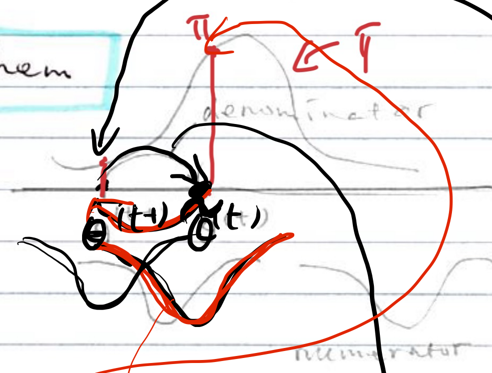
Note that when $x_i \in (y_i, y_i + 1)$

$$I(y_i = \text{floor}(x_i)) = 1$$





⇒ **Metropolis - Hasting Algorithem**



starting with $\theta^{(0)}$

Repeat n times

1) Draw $\tilde{\theta}^{(t)} \sim T(\theta \mid \theta^{(t)})$

2) compute $r = \min\left\{ \dfrac{\pi(\tilde{\theta}^{(t)})\, T(\theta^{(t-1)} \mid \tilde{\theta}^{(t)})}{\pi(\theta^{(t-1)})\, T(\tilde{\theta}^{(t)} \mid \theta^{(t-1)})}, 1 \right\}$

3) Draw $U \sim unif([0, 1])$

if $U < r$

then $\theta^{(t)} = \tilde{\theta}^{(t)}$

else $\theta^{(t)} = \theta^{(t-1)}$

Metropolis - Hasting : starting with $\theta^{(0)}$

Repeat n times

1) draw $\theta^* \sim T(\theta \mid \theta^{(t-1)})$

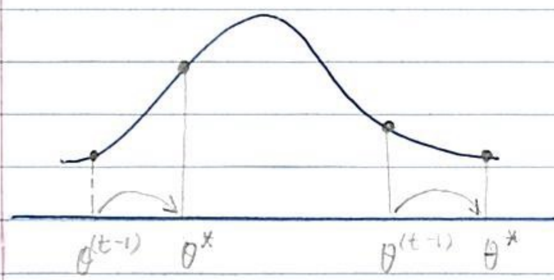
2) compute accept Rate

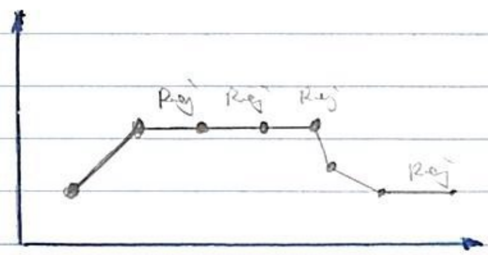
$$r = \min\left\{ \frac{\pi(\theta^*)\, T(\theta^{(t-1)} \mid \theta^*)}{\pi(\theta^{(t-1)})\, T(\theta^* \mid \theta^{(t-1)})},\ 1 \right\}$$

3) draw $U \sim unif([0,1])$

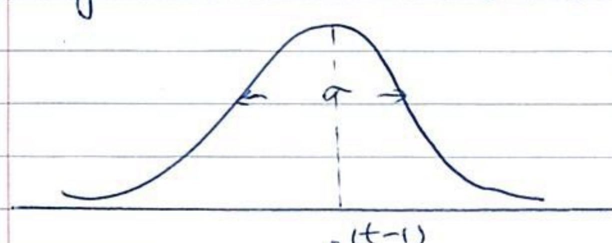
if $(U < r)$ Let $\theta^{(t)} = \theta^*$

else let $\theta^{(t)} = \theta^{(t-1)}$

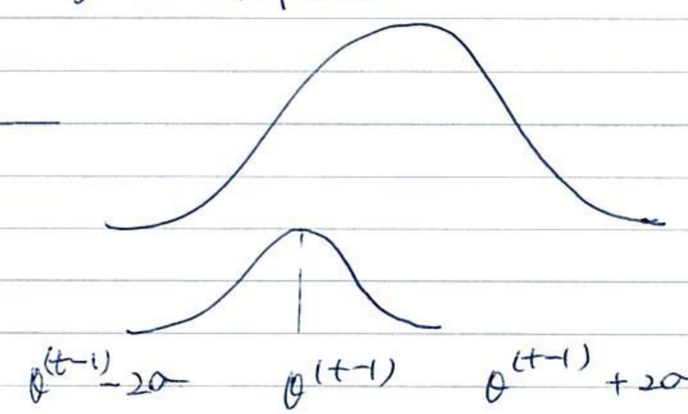




$\Rightarrow$ symetric : $T(\theta \mid \theta^{(t-1)}) = N(\theta^{(t-1)}, \sigma^2)$



σ — stepwise

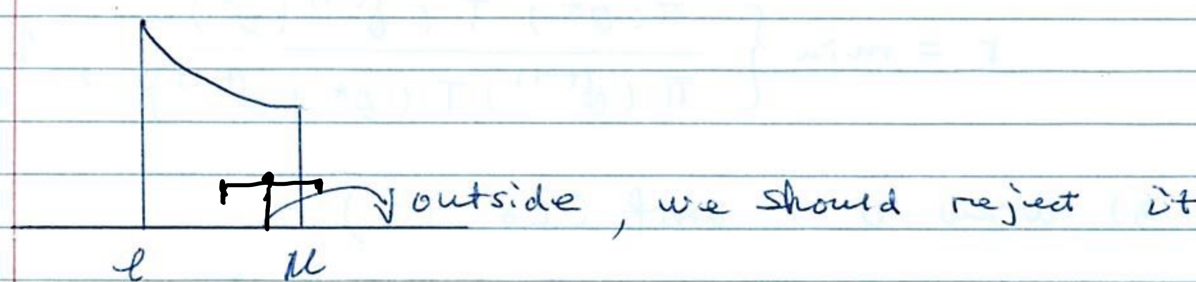


example : truncnormal

$$\pi(\theta) \propto \varphi(\theta)\, I(l \le \theta \le u)$$

Rejection weight :

$$R = \min\left(\frac{\pi(\theta^*)}{\pi(\theta^{(t-1)})} , 1 \right)$$



$\Rightarrow$ 1) optimal accept Rate $= 0.234$

2) optimal Rejection Rate
$= 1 -$ Accept Rate
≈ 0.75

* Note that : Rejection Rate too lower can cause many problems.

* $Var(\bar{x}) = \frac{\sigma^2}{T} (1 + \rho_1 + \rho_2 + \cdots)$

choice of stepsize

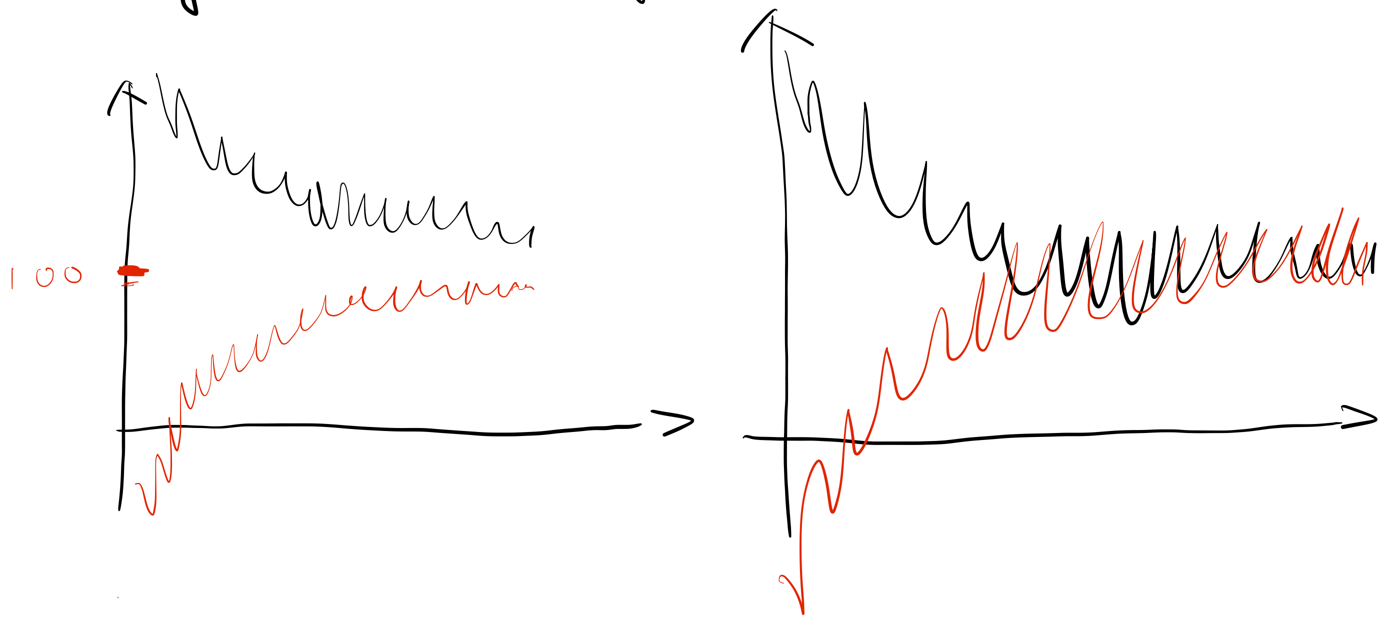
1) optimal accept Rate $= 0.234$

2) optimal Rejection Rate
$= 1 -$ Accept Rate
≈ 0.75

* Note that : Rejection Rate too lower
can cause many problems.

* $Var(\bar{x}) = \frac{\sigma^2}{T} (1 + \rho_1 + \rho_2 + \cdots)$

Diagnostic of Convergence



chain 1: $\theta_1^{(1)}, \cdots, \theta_n^{(1)} \longrightarrow S_{(1)}^2$

chain 2: $\theta_1^{(2)}, \cdots, \theta_n^{(2)} \longrightarrow S_{(2)}^2$

$\vdots$

chain m: $\theta^{(m)}, \cdots, \theta_n^{(m)} \longrightarrow S_{(m)}^2$

$$\hat{R} = \frac{S_T^2}{S_W^2}$$ where

$$S_W^2 = \sum_{j=1}^{m} S_{(j)}^2 / m$$

$$S_T^2 = \sum_i \sum_j \left(\theta_i^{(j)} - \bar{\theta}_{..}\right)^2 / (nm-1)$$

$$\hat{R} = \begin{cases} < 1.05 & : \text{good} \\ \in (1.05, 1.1) & : \text{fair} \\ > 1.1 & : \text{bad} \end{cases}$$

Example: posterior of Normal Sample

$\Rightarrow$ Metropolis – hasting

$X_1, X_2, \cdots, X_n \sim N(\mu, \sigma^2)$

$\mu \sim N(\mu_0, \sigma_0^2)$

$\sigma^2 \sim IG\left(\frac{\alpha}{2}, \frac{\alpha w}{2}\right)$

$P(\mu, \sigma^2) \propto P(X_1, X_2 \cdots X_n \mid \mu, \sigma^2) \cdot P(\mu) \cdot P(\sigma^2) \cdot \sigma^2$

(Let $\log\sigma^2 = y$)

$\propto P(X_1 \cdots, X_n \mid \mu, \sigma^2) P(\mu) P(\sigma^2) \cdot e^{\log\sigma^2}$

$\sigma^2 = e^{\log\sigma^2}$

$w = \log\sigma^2, \quad \sigma^2 = e^{w}$

$\Rightarrow X \sim f_X(x)$

$Y \sim t(x) \Rightarrow X = t^{-1}(Y)$

$$f_Y(y) = f_X(t^{-1}(y)) \left| \frac{\partial t^{-1}(y)}{\partial y} \right|$$

$y = \log\sigma^2 \qquad \sigma^2 = e^{Y} \qquad Y = \log\sigma^2$

> justification of Metropolist Hasting

Review $\boxed{\pi(\theta)\, T(\theta \to \theta^T) = \pi(\theta^T)\, T(\theta^T \to \theta)}$

Metropolist Hasting Algorithm:

From θ

1) Draw $\theta^* \sim \tilde{T}(\cdot \mid \theta^{(t)})$

2) set $\theta^{(t)} = \theta^*$ with

$$r = \min\left\{ \frac{\pi(\theta^*)\, \tilde{T}(\theta^* \to \theta)}{\pi(\theta)\, \tilde{T}(\theta \to \theta^*)},\ 1 \right\}$$

OR, set $\theta^T = \theta$

the $T(\theta \to \theta^T)$ in Metropolist Hasting

* If $\theta = \theta^T$, we do not have

expression $T(\theta \to \theta^T)$ $\quad \underrightarrow{\theta \to \theta^T}$

* If $\theta = \theta^T$, $T(\theta \to \theta^T) = \tilde{T}(\theta \to \theta^T)$

$$\times \min\left\{ \frac{\pi(\theta^T)\, \tilde{T}(\theta^T \to \theta)}{\pi(\theta)\, \tilde{T}(\theta \to \theta^T)},\ 1 \right\}$$

$$\Rightarrow \quad LHS = \pi(\theta)\, T(\theta \to \theta^T)$$

$$= \pi(\theta) \cdot \tilde{T}(\theta \to \theta^T) \cdot \min\left\{ \frac{\pi(\theta^T)\, \tilde{T}(\theta^T \to \theta)}{\cancel{\pi(\theta)\, \tilde{T}(\theta \to \theta^T)}},\ 1 \right\}$$

$$= \min\left(\pi(\theta^T) \cdot \tilde{T}(\theta^T \to \theta),\ \pi(\theta)\, \tilde{T}(\theta \to \theta^T) \right)$$

$$RHS = \min\left(\pi(\theta)\, \tilde{T}(\theta \to \theta^T),\ \pi(\theta^T)\, \tilde{T}(\theta^T \to \theta) \right)$$

⇒ Metroplist - within - Gibbs

$\theta = (\theta_1, \theta_2)$

starting $(\theta_1^{(0)}, \theta_2^{(0)})$

Repeat :

1) Draw $\theta_1^{(t)} \sim \pi(\theta_1 \mid \theta_2^{(t-1)})$

2) Draw $\theta_2^{(t)} \sim \pi(\theta_2 \mid \theta_1^{(t-1)})$

In any steps, say for θ_i, in Gibbs

We could replace the direct sampling

by Metroplist hasting transition $T_i(\theta_i \to \theta_i^T)$

that leaves $\pi(\theta_i \mid \theta_{-i}^{(t-1)})$ invariant,

In other words, we apply M - H to

the target distribution $\pi(\theta_i \mid \theta_{-i})$

WinBugs,

Bugs means Bayesian Inference Using Gibbs

Jags : it is just another Gibbs Sampler

Downlow Jags

install.packages("R2jags")

Library("R2jags")

example : $X_1, \dots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$

$$\mu \sim N(\mu_0, \sigma_0^2)$$

$$\sigma^2 \sim IG\left(\frac{\alpha}{2}, \frac{\alpha w}{2}\right)$$

Go to file page -> menue

Note that : $X_1, \dots X_n \overset{iid}{\sim} N(\mu, \tau)$

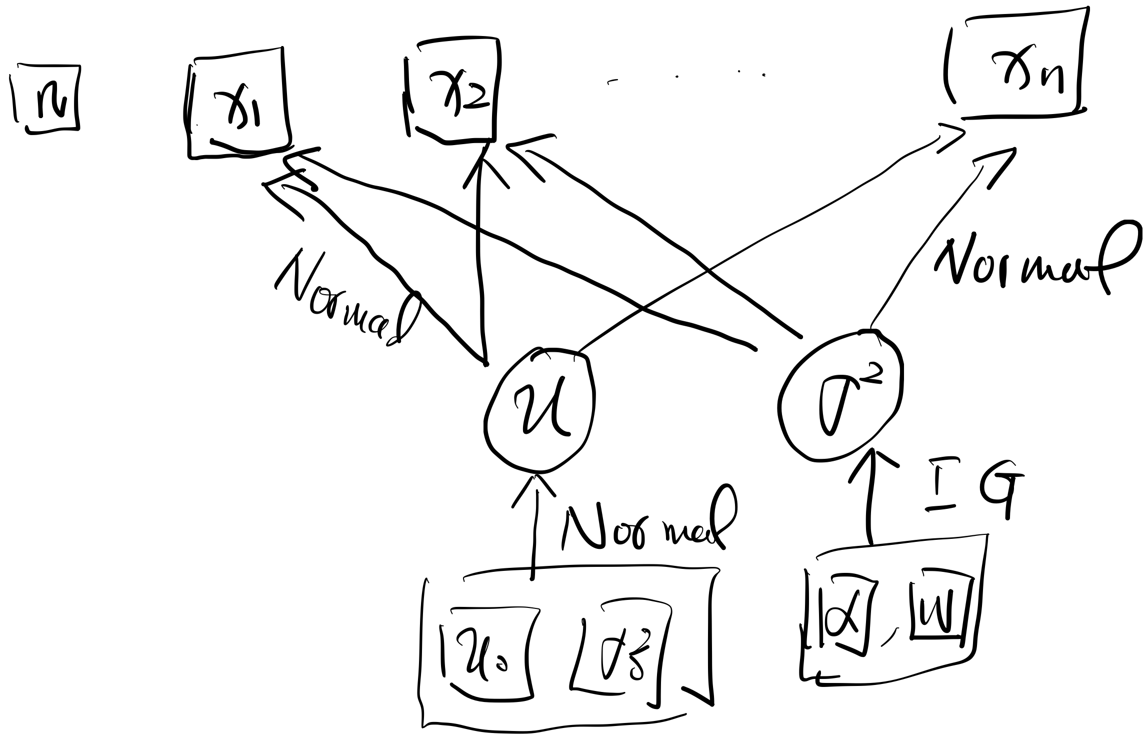
$$\mu \sim N(\mu_0 \ \sigma_0^2)$$

$$\tau \sim \text{Gamma}\left(\frac{\alpha}{2}, \frac{\alpha w}{2}\right)$$

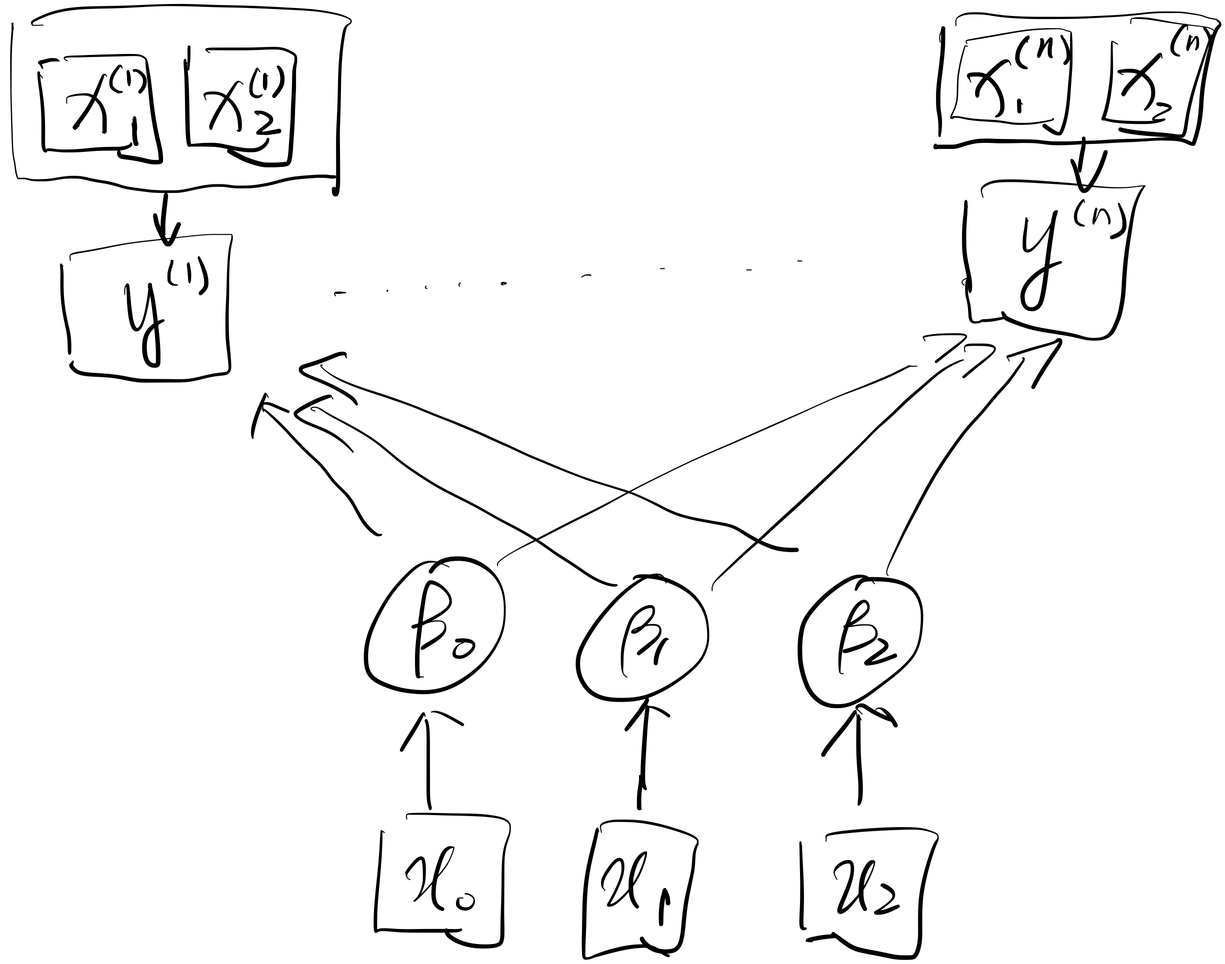
wher $\tau = 1/\sigma^2$, τ dons't means variance. onty means precision

$$p(\mu, \sigma^2 \mid x_1, \ldots, x_n)$$

$$\propto \prod_{i=1}^{n} f(x_i \mid \mu, \sigma^2) \cdot \pi(\mu)\pi(\sigma^2)$$



Example: Logistic Regression



$$p^{(i)} = \frac{1}{1 + e^{-[\beta_0 + \beta_1 x_1^{(i)} + \beta_2 x_2^{(i)}]}}$$

$$\log\left(\frac{p^{(i)}}{1-p^{(i)}}\right) = \beta_0 + \beta_1 x_1^{(i)} + \beta_2 x_2^{(i)}$$